# Distribution Shift, OOD Detection, Adversarial Robustness, and Conformal Prediction

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**Abstract**

This document is a *layered tutorial survey*. It is written for readers who know basic linear algebra, probability, and calculus but may be new to reliability problems that arise when machine-learning systems leave the training distribution. The survey covers four tightly connected topics: *distribution shift*, *out-of-distribution (OOD) detection*, *adversarial robustness*, and *conformal prediction under shift*. The central organizing idea is simple: a model is trained on a source joint distribution $P_{\mathrm{s}}(X, Y)$ and deployed on a target joint distribution $P_{\mathrm{t}}(X, Y)$; the right response depends on *what changed*. If only the covariates shift, reweighting may be enough. If class priors shift, posterior correction may be enough. If new semantics appear, the model should often reject rather than force a label. If an attacker can perturb inputs within a threat model, the problem becomes worst-case optimization. If the goal is reliable set-valued uncertainty rather than a single label, conformal prediction becomes the right tool.

The document is designed in four layers. **Part I** is the guided core path: it introduces a running example, a symbol table, micro-prerequisites with tiny worked examples, a one-page taxonomy based on the factorization $p(x, y) = p(y \mid x)p(x) = p(x \mid y)p(y)$, and plain-language explanations of the four frameworks. **Part II** gives detailed mathematical derivations for the canonical frameworks: importance weighting for covariate shift; posterior correction and confusion-matrix inversion for label shift; score-threshold OOD detection with MSP, energy, and Mahalanobis scores; adversarial robustness through the min–max objective, FGSM, PGD, adversarial training, TRADES, and a simple certified-robustness derivation via interval bound propagation; and conformal prediction through split conformal, APS, RAPS, weighted conformal, label-shift conformal, and adaptive online conformal methods. Each derivation is organized as *goal* $\to$ *assumption* $\to$ *mathematical move* $\to$ *final algorithm*. **Part III** is the survey layer: it maps historically important works from roots to leaves, and then reviews recent influential papers from roughly 2023–2025, explaining how each one modifies the canonical framework, what it claims, and what evidence it uses to support those claims. **Part IV** gives practical checklists, reporting conventions, and open problems.

Historically important works covered here include Shimodaira on covariate shift, Saerens-style prior correction, BBSE for label shift, Quionero-Candela and Moreno-Torres on dataset shift, Ben-David et al. on domain adaptation, Vovk–Gammerman–Shafer on conformal prediction, Bendale and Boult on OpenMax, Hendrycks and Gimpel on the maximum-softmax baseline, ODIN, Mahalanobis OOD, Outlier Exposure, Szegedy et al. on adversarial examples, Goodfellow et al. on FGSM, Madry et al. on adversarial training, TRADES, AutoAttack, covariate-shift conformal prediction, and label-shift conformal prediction. Recent influential works covered here include NPOS, SAL, NINCO, ImageNet-OOD, OpenOOD v1.5, and the 2025 OOD-X / real-world

VLM benchmark line on the OOD side; OODRobustBench, AdvXL, Robust CLIP, subset-based adversarial training, RAMP, and conflict-aware adversarial training on the adversarial side; and the recent tutorial/book line on conformal prediction, conformal PID control, online conformal inference under arbitrary shift, unlabeled target adaptation, multi-source conformal inference, and error-quantified conformal inference on the conformal side.

# Contents

**How to use this document**

This survey is intentionally split into a **core path** and a **survey path**.

- **If you are new to the topic, read** Part I, then Sections 6–9 in Part II, and only then continue to Parts III–IV.
- **If you already know the basics,** jump directly to Part II for full derivations.
- **If you mainly want the literature survey,** read Part III after skimming the one-page cheat sheet in Section 1.
- Gray boxes are **optional advanced notes**. They can be skipped on a first reading without losing the main story.

Each major section follows the same pattern:

deployment problem → assumption → quantity we compute → final rule → failure mode.

## Part I

# Part I: Core tutorial path

## 1 Big picture, running example, and micro-prerequisites

### 1.1 The single question behind this survey

A model almost never lives forever inside the exact training distribution. Training data come from one environment, deployment data come from another, and reliability problems appear because the two environments differ. The simplest mathematical way to say this is

$$P_{\mathrm{s}}(X, Y) \neq P_{\mathrm{t}}(X, Y),$$

where $P_{\mathrm{s}}$ is the *source* (training) joint distribution and $P_{\mathrm{t}}$ is the *target* (deployment) joint distribution.

The entire survey is built around one deployment question:

*When the world changes at test time, should I reweight the loss, reject the input, defend against worst-case perturbations, or return a larger uncertainty set?*

Those are four different responses to four different kinds of mismatch.

**One-page cheat sheet**

| **Topic** | Deployment question | Main assumption | Quantity we compute | Final action |
|---|---|---|---|---|
| **Distribution shift** | The task is the same, but the data distribution changed | Some part of the data-generating mechanism stays stable | A target-risk estimate or a corrected posterior | Keep predicting, but correct for the shift |
| **OOD detection** | The input may belong to no training class | ID inputs should score higher than OOD inputs under some detector score | A scalar score and a threshold | Accept or reject the input |
| **Adversarial robustness** | An attacker can perturb the input within a budget | Threat model is specified in advance | Worst-case loss inside a perturbation set | Train/evaluate against the worst case |
| **Conformal prediction** | We want reliable uncertainty sets, not only a point prediction | Exchangeability, or a carefully adapted version under shift | A score quantile | Output a prediction set or interval |

## 1.2 A running example used throughout the document

To keep the story concrete, we use one running example.

A classifier is trained to recognize $\{\text{cat}, \text{dog}, \text{rabbit}\}$ from clean studio-style pet photos. At deployment the classifier is used by an animal shelter. Several things can go wrong:

- The shelter photos are darker and have cluttered backgrounds. The input distribution changed, even though the label set stayed the same.
- Dogs are much more common in the shelter than in the training data. Class priors changed.
- The shelter also receives hamsters and foxes, which were not in the training classes. New semantics appeared.
- A malicious user adds a tiny adversarial sticker or pixel-level perturbation that changes the prediction while keeping the image visually almost unchanged.
- Instead of a single label, the shelter wants the model to return a reliable set such as $\{\text{dog}, \text{rabbit}\}$ when it is unsure.

These are exactly the four topics of the survey.

## 1.3 The factorization that explains most of the taxonomy

A beginner does not need dozens of definitions at the start. Most of the basic taxonomy comes from one identity:

$$p(x, y) = p(y \mid x)p(x) = p(x \mid y)p(y).$$

This identity says that a joint distribution can be viewed from two angles.

- $p(x)$ tells us how inputs are distributed.
- $p(y)$ tells us how class frequencies are distributed.
- $p(y \mid x)$ tells us how labels depend on inputs.
- $p(x \mid y)$ tells us how inputs look inside each class.

So when source and target differ, a first beginner question is:

*Which factor changed?*

That single question already points toward the right framework.

## 1.4 Micro-prerequisites with tiny worked examples

A true novice does not only need names of prerequisites; they need tiny examples. This subsection gives the only mathematics you need before reading the main story.

**Prerequisite 1: Bayes' rule**

Bayes' rule says

$$p(y \mid x) = \frac{p(x \mid y)p(y)}{p(x)}.$$

If an image $x$ looks equally compatible with cat and dog inside each class, but dogs are twice as common as cats in deployment, then the posterior should tilt toward dog because $p(y)$ changed. This is the basic idea behind *label-shift correction*.

**Prerequisite 2: risk = average loss**

If $f$ is a classifier and $\ell(f(x), y)$ is its loss on $(x, y)$, then its risk on a distribution $P$ is

$$R_P(f) = \mathbb{E}_{(X,Y)\sim P}\big[\ell(f(X), Y)\big].$$

If the deployment distribution changes, the number we care about is $R_{P_t}(f)$, not $R_{P_s}(f)$. Distribution-shift methods try to estimate or control this target risk.

**Prerequisite 3: a gradient points toward local increase**

If $g(x)$ is differentiable, then for a small perturbation $\delta$,

$$g(x + \delta) \approx g(x) + \nabla g(x)^\top \delta.$$

So if we want to increase a loss quickly, we move $\delta$ in a direction aligned with the gradient. This is the seed from which FGSM and PGD grow.

**Prerequisite 4: norms define perturbation budgets**

The set

$$\{\delta : \|\delta\|_\infty \leq \varepsilon\}$$

means every coordinate can move up or down by at most $\varepsilon$. This is why $\ell_\infty$ attacks are often described as "small changes to every pixel." By contrast, $\ell_2$ controls overall Euclidean size.

**Prerequisite 5: a quantile is a threshold**

If calibration scores are

$$0.10,\ 0.18,\ 0.22,\ 0.31,\ 0.45,$$

then the 80% quantile is the smallest value such that at least 80% of the scores are below it. In this tiny list, that threshold is 0.31 if we use the usual finite-sample conformal indexing rule. Conformal prediction works by comparing a new score with such a threshold.

**Prerequisite 6: exchangeability is "order should not matter"**

If calibration examples and the test example are exchangeable, then after we compute their scores, the rank of the test score among all scores is uniformly distributed over all possible rank positions. This simple symmetry fact is the engine behind split conformal coverage.

### 1.5 A local symbol table

| **Symbol** | Meaning | Plain-language interpretation |
|---|---|---|
| $P_{\mathrm{s}}, P_{\mathrm{t}}$ | source and target distributions | training world and deployment world |
| $X, Y$ | input and label | image and class name |
| $f$ | model or predictor | the learned system we deploy |
| $\ell(f(x), y)$ | loss | how wrong the prediction is on one example |
| $R_P(f)$ | risk under $P$ | average loss on a distribution |
| $w(x)$ | importance weight | how much target world values input $x$ relative to source world |
| $r(y)$ | label-prior ratio | how much more/less common class $y$ is at test time |
| $S(x)$ | OOD score | larger means "more ID-like" unless stated otherwise |
| $\tau$ | threshold | cut-off used for accept/reject decisions |
| $\delta$ | perturbation | small input change used by an adversary |
| $\varepsilon$ | perturbation budget | maximum allowed perturbation size |
| $V(x, y)$ | conformal score | how incompatible label $y$ looks for input $x$ |
| $\hat{q}$ | calibrated threshold | score quantile used to form prediction sets |

**Common pitfall**

Many beginners treat all test-time problems as "distribution shift." That phrase is too broad. A change in lighting, a new class, an adversarial perturbation, and a need for valid uncertainty sets are *not* the same problem. The rest of the document is about learning to tell them apart.

**What to remember**

Before reading anything else, remember this: reliability under shift starts by asking *what changed*. If the answer is "the same classes, but different input frequencies," think about covariate or label shift. If the answer is "new semantic classes may appear," think about OOD detection. If the answer is "an attacker can choose perturbations," think about adversarial robustness. If the answer is "I need formal set-valued uncertainty," think about conformal prediction.

## 2 Shift taxonomy: what changed between source and target?

## 2.1 Practical map of the taxonomy

**Summary: shift taxonomy**

**Problem.** The source and target distributions are not the same.
**Question.** Which part of $p(x, y) = p(y \mid x)p(x) = p(x \mid y)p(y)$ changed?
**Most common cases.**

- **Covariate shift:** $p_{\mathrm{t}}(x) \neq p_{\mathrm{s}}(x)$, but $p_{\mathrm{t}}(y \mid x) = p_{\mathrm{s}}(y \mid x)$.
- **Label shift:** $p_{\mathrm{t}}(y) \neq p_{\mathrm{s}}(y)$, but $p_{\mathrm{t}}(x \mid y) = p_{\mathrm{s}}(x \mid y)$.
- **Semantic shift:** the training label set is not enough at deployment; new or reorganized meanings/classes appear.

**Why it matters.** These cases require different fixes.

## 2.2 Covariate shift

Covariate shift means the target environment shows the model different kinds of inputs, but the *correct answer for a given input* does not change:

$$p_{\mathrm{t}}(x) \neq p_{\mathrm{s}}(x), \qquad p_{\mathrm{t}}(y \mid x) = p_{\mathrm{s}}(y \mid x).$$

In the shelter example, this means the camera, lighting, background, or pose distribution changes, but an image that truly depicts a dog is still a dog. The task itself is unchanged.

**Toy example**

Suppose training images are mostly clean front-facing pet portraits, while shelter images are taken in dim light from many angles. The distribution of $X$ changed. But if two images are visually identical, the correct label should still be the same in training and deployment. That is the intuition behind $p_{\mathrm{t}}(y \mid x) = p_{\mathrm{s}}(y \mid x)$.

Why is this special? Because if $p(y \mid x)$ is stable, then target risk can often be rewritten as a weighted source expectation. That is the key mathematical move behind importance weighting, developed in Section 6.

## 2.3 Label shift

Label shift means class frequencies change, but the class-conditional appearance stays stable:

$$p_{\mathrm{t}}(y) \neq p_{\mathrm{s}}(y), \qquad p_{\mathrm{t}}(x \mid y) = p_{\mathrm{s}}(x \mid y).$$

In the shelter example, perhaps dogs are much more common than rabbits, even though the visual distribution of dogs and rabbits *within their classes* has not changed.

**Toy example**

Imagine the model was trained on a balanced dataset with one third cats, one third dogs, and one third rabbits. At deployment the shelter has 60% dogs, 25% cats, and 15% rabbits. The model's posterior probabilities should shift even if the visual appearance of each class stays the same.

This case is important because Bayes' rule tells us that changing priors $p(y)$ changes posteriors $p(y \mid x)$. That is why posterior correction is the main tool here.

## 2.4 Semantic shift

A closely related case is *concept* or *conditional* shift:

$$p_{\mathrm{t}}(y \mid x) \neq p_{\mathrm{s}}(y \mid x).$$

This means the correct label rule itself changed. In that case, simple covariate-shift weighting or label-shift correction is usually not enough. We mention this explicitly because students often ask where it fits. In this survey, the user-requested taxonomy emphasizes *covariate shift*, *label shift*, and *semantic shift*. We use *semantic shift* for the practically important case where the deployment semantics are not captured by the closed training label set.

In this survey, *semantic shift* means the deployment semantics are not captured by the closed training label set. Examples include:

- new classes appear (hamster, fox);
- the test hierarchy is finer or coarser than the training hierarchy;
- the deployment question itself changes so the training labels are no longer the right partition of the world.

This is different from covariate shift or label shift. If a fox appears at test time, the problem is not that the same old classes have different frequencies. The problem is that the model is being asked to operate outside its closed-world semantic assumptions.

That is why OOD detection, open-set recognition, or abstention becomes necessary.

## 2.5 Near-OOD versus far-OOD

Not all OOD inputs are equally difficult.

- **Near-OOD** examples are semantically outside the training classes but visually or structurally similar to them. In the pet example, a *fox* is near-OOD because it can resemble dog-like features.
- **Far-OOD** examples are far from the training manifold in both appearance and semantics. In the pet example, an *airplane* image is far-OOD.

Near-OOD is usually harder than far-OOD. A detector can often reject airplanes easily, but foxes may still receive high dog scores because the representation overlaps.

**Common pitfall**

A corrupted dog photo is usually *not* OOD in the semantic sense. It is more naturally described as a *shifted in-distribution example* or a covariate shift example. Beginners often mix these cases.

## 2.6 Dataset construction pitfalls

A large fraction of the OOD and shift literature is about avoiding bad benchmarks. A novice should know these pitfalls early because they affect how to interpret almost every reported number.

| **Pitfall** | Why it breaks interpretation | Better practice |
|---|---|---|
| **Preprocessing mismatch** | The detector may exploit image size, compression, or normalization artifacts instead of semantics | Match preprocessing carefully across ID and OOD datasets |
| **Background shortcuts** | The model detects background or texture bias rather than true semantic novelty | Curate OOD sets that reduce shortcut cues |
| **ID/OOD contamination** | Overlapping classes or near duplicates inflate or deflate performance | Remove overlap, duplicates, and hierarchical collisions |
| **Mixing shift types** | One benchmark can accidentally combine semantic shift with covariate artifacts | Separate semantic shift and covariate shift when possible |
| **Threshold tuning on the test OOD set** | Reported thresholds no longer reflect deployment reality | Tune on validation data or report threshold-free metrics as well |
| **Overly easy far-OOD only** | Methods look strong because the benchmark contains obvious outliers | Include near-OOD and realistic "shifted ID" cases |

Recent papers such as NINCO, ImageNet-OOD, OpenOOD v1.5, and the 2025 OOD-X benchmark family became influential largely because they improved the *evaluation protocol*, not only the detector score [32, 38, 39, 49].

**What to remember**

For a beginner, the practical taxonomy is:

- if the same examples would get the same correct labels but appear in different proportions, think **covariate shift**;
- if class frequencies changed, think **label shift**;

- if new classes or meanings appear, think **semantic shift / OOD**;
- if worst-case perturbations matter, think **adversarial robustness**;
- if you need a formal uncertainty set, think **conformal prediction**.

# 3 OOD detection: one score and one threshold

## 3.1 The whole idea in one sentence

**Summary: OOD detection**

**Problem.** The input may not belong to any training class.
**Assumption.** In-distribution inputs should look more "typical" than OOD inputs according to some score.
**Compute.** A scalar score $S(x)$.
**Decision rule.** Accept as ID if $S(x) \geq \tau$, otherwise reject as OOD.
**Failure mode.** A score may be high for the wrong reason, for example because it reacts to texture, logit scale, or background shortcuts rather than true semantics.

That is the core idea. Different OOD methods mostly differ in *how they compute* $S(x)$.

## 3.2 Maximum softmax probability (MSP)

Suppose the classifier outputs logits $z_1(x), \ldots, z_K(x)$. The softmax probabilities are

$$p_\theta(y \mid x) = \frac{e^{z_y(x)}}{\sum_{k=1}^{K} e^{z_k(x)}}.$$

The simplest OOD score is

$$S_{\mathrm{MSP}}(x) = \max_y p_\theta(y \mid x).$$

If the model is very confident in one class, $S_{\mathrm{MSP}}(x)$ is large. If the probability mass is spread more evenly, $S_{\mathrm{MSP}}(x)$ is smaller.

**Toy example**

Consider three logit vectors for a three-class model.

- $z = (6, 1, 0)$. Softmax is highly peaked, so MSP is close to 1.
- $z = (2, 2, 2)$. Softmax is uniform, so MSP is 1/3.
- $z = (10, 9, 8)$. The model still has a preferred class, but the relative gaps are modest. MSP is much lower than in the first example even though all logits are large in absolute value.

MSP only sees *relative probabilities*; it does not care about overall logit scale.

MSP became historically important because it is simple, strong, and hard to beat in some settings [11]. It is still the baseline every new method must surpass.

### 3.3 Energy scores

MSP can hide absolute logit scale. Energy scores try to use that scale:

$$S_{\text{energy}}(x) = T \log \sum_{y=1}^{K} e^{z_y(x)/T},$$

where $T > 0$ is a temperature parameter. Some papers define $-S_{\text{energy}}$ as an energy to be minimized; throughout this survey we treat larger $S_{\text{energy}}$ as "more ID-like."

Why can this help? Because log-sum-exp gets larger when *all* logits shift upward. MSP cannot see that.

**Toy example**

Use the same examples as above.

- For $z = (6, 1, 0)$, both MSP and energy are high.
- For $z = (2, 2, 2)$, MSP is low and energy is moderate.
- For $z = (10, 9, 8)$, MSP may still be only moderate, but energy is very high because all logits are large.

This is why energy can separate some ID and OOD examples that MSP treats similarly.

### 3.4 Mahalanobis-style scores

Mahalanobis methods step away from output probabilities and look inside the feature space. Let $\phi(x) \in \mathbb{R}^d$ be the feature representation from some internal layer. For class $y$, estimate a feature mean $\mu_y$ and a shared covariance matrix $\Sigma$. Then define

$$S_{\text{Mah}}(x) = -\min_y (\phi(x) - \mu_y)^\top \Sigma^{-1} (\phi(x) - \mu_y).$$

The idea is simple: an ID example should land near at least one class cluster in feature space. An OOD example should land far from all clusters.

**Toy example**

Suppose dog features vary a lot along one direction (say pose) but very little along another direction (say ear shape). Plain Euclidean distance treats both directions equally. Mahalanobis distance rescales directions using $\Sigma^{-1}$, so being far in a low-variance direction counts more heavily. That is why it is often better than raw Euclidean distance.

### 3.5 Thresholding and calibration

After choosing a score, a detector still needs a threshold $\tau$. A novice should separate three ideas:

- **The score:** how we map an input to one number.
- **The threshold:** where we cut between accept and reject.
- **Calibration protocol:** which data we use to choose that threshold.

In deployment, the threshold should usually be chosen on a validation set, or through a reporting convention such as fixing a target true-positive rate on ID data and measuring the resulting false-positive rate on OOD data. If a paper tunes $\tau$ directly on the test OOD benchmark, it is no longer evaluating a realistic deployment setting.

### 3.6 Why a beginner should already care about ODIN and Outlier Exposure

Even before the detailed derivations in Part II, it is useful to know two historically important improvements:

- **ODIN** sharpens the softmax output using temperature scaling and adds a tiny input perturbation chosen to increase confidence [13].
- **Outlier Exposure (OE)** trains with auxiliary outlier data so the model learns to assign low confidence to things it should reject [18].

Both still fit the same beginner template: *produce a score, then threshold it.* They simply modify the score or the training procedure.

**Common pitfall**

Do not think of OOD detection as "the classifier gives the wrong label." A classifier can be wrong *within* the training classes. OOD detection is about deciding whether the input should have been forced into the training label set in the first place.

**What to remember**

A novice should leave this section remembering only three things:

1. OOD detection is mostly **score + threshold**.
2. MSP, energy, and Mahalanobis are the three most common score families.
3. Good evaluation matters as much as good scoring, because shortcuts in the benchmark can completely change which method looks best.

## 4 Adversarial robustness: when worst-case perturbations matter

### 4.1 Threat models come first

**Summary: adversarial robustness**

**Problem.** An attacker can perturb the input to increase loss or change the prediction.
**Assumption.** We know the allowed perturbation set, for example $\|\delta\|_\infty \le \varepsilon$.
**Compute.** The worst-case loss inside that set.
**Decision rule.** Train or evaluate with respect to the worst case, not only the clean input.
**Failure mode.** Robustness claims are meaningless if the threat model or evaluation attack is weak or unspecified.

Before any attack or defense, we must specify a *threat model*. A threat model answers at least four questions:

- What can the attacker modify?
- How large can the modification be?
- Does the attacker know the model?
- Is the goal targeted (force a specific wrong class) or untargeted (any wrong class)?

Without a threat model, the phrase "robust model" is incomplete.

### 4.2 FGSM in the simplest $\ell_\infty$ picture

The fast gradient sign method (FGSM) is the first attack every beginner should understand [8]. Let the loss be $\ell(f_\theta(x), y)$. Around a given input $x$, linearize the loss:

$$\ell(f_\theta(x+\delta), y) \approx \ell(f_\theta(x), y) + \nabla_x \ell(f_\theta(x), y)^\top \delta.$$

Under an $\ell_\infty$ budget, each coordinate of $\delta$ can move independently between $-\varepsilon$ and $+\varepsilon$. To maximize the inner product, choose each coordinate to have the sign of the corresponding gradient coordinate:

$$\delta_{\mathrm{FGSM}} = \varepsilon \ \mathrm{sign}(\nabla_x \ell(f_\theta(x), y)) \, .$$

The adversarial example is

$$x_{\mathrm{adv}} = x + \delta_{\mathrm{FGSM}}.$$

**Toy example**

If the gradient with respect to three pixels is $(2.1, -0.3, 5.0)$, then under $\varepsilon = 0.01$ the FGSM perturbation is

$$(0.01, -0.01, 0.01).$$

FGSM moves each pixel in the direction that increases loss as quickly as possible under the coordinate-wise budget.

### 4.3 PGD: repeated FGSM with projection

FGSM is a one-step attack. Projected gradient descent (PGD) takes many small steps:

$$x^{t+1} = \Pi_{B_\varepsilon(x)}\Big(x^t + \alpha \ \mathrm{sign}(\nabla_{x^t} \ell(f_\theta(x^t), y))\Big),$$

where $B_\varepsilon(x)$ is the allowed perturbation ball and $\Pi$ projects back into it.

A good beginner picture is: FGSM is one large push; PGD is many small pushes, each followed by "snap back into the allowed region."

PGD became central because a strong first-order attack is needed both for evaluation and for adversarial training [51].

### 4.4 What adversarial training does

Ordinary training minimizes average clean loss:

$$\min_\theta \mathbb{E}[\ell(f_\theta(X), Y)].$$

Adversarial training replaces the clean loss with the *worst* loss inside a perturbation set:

$$\min_\theta \mathbb{E}\Big[\max_{\|\delta\| \le \varepsilon} \ell(f_\theta(X + \delta), Y)\Big].$$

So the model is trained against the attack we fear. This is the min–max framework that dominates adversarial robustness.

### 4.5 Why robustness and clean accuracy can conflict

Robustness often requires smoother or larger-margin decision boundaries. That can reduce clean accuracy, especially when the data contain fragile but useful features. This tension was emphasized by the robust-accuracy tradeoff literature [20, 21].

A beginner-friendly way to think about it is:

*A model that squeezes every last bit of predictive signal from tiny high-frequency cues may classify clean data well, but those same cues may be easy for an adversary to manipulate.*

TRADES, discussed in detail later, is a canonical method for managing this tension.

### 4.6 Evaluation basics every beginner should know

A robustness claim should report at least the following:

- perturbation norm and budget, e.g. $\ell_\infty$ with $\varepsilon = 8/255$;

- attack type and strength, e.g. PGD steps, step size, restarts;
- clean accuracy and robust accuracy;
- whether evaluation includes strong attack suites such as AutoAttack [24];
- whether the reported robustness survives distribution shift, not just the original benchmark [44].

**Common pitfall**

A weak attack can make a model look robust even when it is not. This is why the robustness literature repeatedly warns about gradient masking or obfuscated gradients [16].

**What to remember**

The beginner's mental model is:

$$\text{clean loss} \quad \longrightarrow \quad \text{worst-case loss in a perturbation set.}$$

FGSM is the simplest one-step approximation to the inner maximization, PGD is the stronger multi-step version, and adversarial training uses these attacks during learning.

# 5 Conformal prediction: reliable sets instead of single guesses

## 5.1 What conformal prediction tries to guarantee

**Summary: conformal prediction**

**Problem.** We want a prediction set $C(x)$ that contains the true label with probability at least $1-\alpha$.
**Assumption.** Source calibration data and the test point are exchangeable, or we have a principled adaptation under shift.
**Compute.** A nonconformity score and a calibrated quantile $\hat{q}$.
**Decision rule.** Output all labels whose score is at most $\hat{q}$.
**Failure mode.** If exchangeability is broken and no adaptation is used, nominal coverage may fail.

A point classifier must always commit to one label. A conformal predictor is allowed to say

$$C(x) = \{\text{dog}, \text{rabbit}\},$$

meaning "I am not comfortable narrowing it further while preserving coverage."

### 5.2 A full worked split-conformal example

To make the mechanics concrete, suppose we have a classifier and a separate calibration set of $n = 9$ labeled examples. For a candidate label $y$, define the simple nonconformity score

$$V(x, y) = 1 - \hat{p}(y \mid x),$$

so smaller scores mean "more compatible."

Assume the nine calibration scores for the *true* labels are

$$0.10,\ 0.15,\ 0.20,\ 0.30,\ 0.35,\ 0.40,\ 0.55,\ 0.70,\ 0.80.$$

Let the target miscoverage be $\alpha = 0.2$. Split conformal uses the index

$$k = \lceil (n + 1)(1 - \alpha) \rceil = \lceil 10 \times 0.8 \rceil = 8.$$

So the threshold is the 8th smallest calibration score:

$$\hat{q} = 0.70.$$

Now take a new test image and suppose the classifier outputs

$$\hat{p}(\text{dog} \mid x) = 0.45, \qquad \hat{p}(\text{cat} \mid x) = 0.35, \qquad \hat{p}(\text{rabbit} \mid x) = 0.20.$$

Then the candidate-label scores are

$$V(x, \text{dog}) = 0.55, \qquad V(x, \text{cat}) = 0.65, \qquad V(x, \text{rabbit}) = 0.80.$$

Include every label with score at most $\hat{q} = 0.70$. Therefore

$$C(x) = \{\text{dog}, \text{cat}\}.$$

This is the full conformal recipe in miniature:

$$\text{score} \to \text{calibration quantile} \to \text{prediction set}.$$

### 5.3 Why the coverage guarantee appears

If the calibration examples and the test example are exchangeable, then the test score has no privileged rank among the $n + 1$ scores. So the chance that the test score falls above the chosen quantile is at most $\alpha$. This gives finite-sample marginal coverage.

The proof in Part II will make this exact, but the beginner intuition is just "the test score behaves like one more random score from the same bag."

### 5.4 APS and RAPS for classification

For image classification, the simple score $1 - \hat{p}(y \mid x)$ can be crude. Two important classification-set methods are APS and RAPS [22, 25].

First sort labels by predicted probability:

$$\hat{p}_{(1)}(x) \geq \hat{p}_{(2)}(x) \geq \cdots \geq \hat{p}_{(K)}(x).$$

If label $y$ has rank $r_y(x)$, then APS uses

$$V_{\mathrm{APS}}(x, y) = \sum_{j=1}^{r_y(x)} \hat{p}_{(j)}(x).$$

So APS asks: how much cumulative probability must I collect before I reach label $y$?

**Toy example**

Suppose the probabilities are

$$(0.45, 0.35, 0.20).$$

Then the cumulative sums are

$$0.45, \qquad 0.80, \qquad 1.00.$$

If the conformal threshold is 0.78, APS includes the first two labels but not the third. So the prediction set is the smallest probability prefix whose cumulative mass exceeds the threshold.

RAPS adds a small penalty for going too far down the ranking, which often shrinks sets without hurting coverage much in practice.

### 5.5 What shift breaks, and why weighted/adaptive conformal is needed

Split conformal is exact under exchangeability. Under shift, exchangeability can fail. Two common fixes are:

- **Weighted conformal:** if we know or estimate how much more representative each calibration point is of the target environment, we weight calibration scores accordingly.
- **Adaptive online conformal:** if data arrive over time and the distribution drifts, we update the threshold sequentially based on recent errors.

The key beginner idea is simple: ordinary conformal treats every calibration point as equally representative of the test point. Shift-aware conformal relaxes that assumption.

### 5.6 Reporting conventions

When a paper reports conformal performance under shift, a beginner should expect at least:

- **coverage:** empirical frequency that the true label lies in the set;
- **average set size** (classification) or interval length (regression);
- **coverage stratified by subgroup or shift type** when possible;
- **the exact score used:** softmax score, APS, RAPS, residual, etc.;
- **the adaptation protocol:** ordinary split conformal, weighted conformal, label-shift conformal, ACI, ECP, and so on.

**What to remember**

A conformal method always has the same skeleton:

choose a score → calibrate a quantile → include labels whose score is below the threshold.

The hard part under shift is deciding which calibration scores should count, and how much.

**Part II**

# Part II: Canonical frameworks and detailed derivations

## 6 Distribution shift: from target risk to workable algorithms

### 6.1 Common setup

Let $f$ be a predictor and $\ell(f(x), y)$ a loss. The quantity we truly care about at deployment is

$$R_{\mathrm{t}}(f) = \mathbb{E}_{(X,Y)\sim P_{\mathrm{t}}}[\ell(f(X), Y)].$$

The challenge is that we usually have labeled data from $P_{\mathrm{s}}$, not from $P_{\mathrm{t}}$.

So the mathematical question is:

*Can I rewrite $R_{\mathrm{t}}(f)$ using quantities I can estimate from source data and a manageable description of the shift?*

That question generates the standard shift-correction methods.

## 6.2 Covariate shift: full derivation of importance weighting

**Derivation scaffold: covariate shift**

**Goal.** Rewrite target risk using source data.
**Assumption.** $p_{\mathrm{t}}(y \mid x) = p_{\mathrm{s}}(y \mid x)$.
**Move.** Multiply and divide by $p_{\mathrm{s}}(x)$.
**Result.** Target risk becomes a weighted source expectation with weight $w(x) = p_{\mathrm{t}}(x)/p_{\mathrm{s}}(x)$.

Start from the target risk:

$$R_{\mathrm{t}}(f) = \int \ell(f(x), y)\, p_{\mathrm{t}}(x, y)\, dx\, dy.$$

Under covariate shift,

$$p_{\mathrm{t}}(x, y) = p_{\mathrm{t}}(x)\, p_{\mathrm{t}}(y \mid x) = p_{\mathrm{t}}(x)\, p_{\mathrm{s}}(y \mid x).$$

Insert this into the risk:

$$R_{\mathrm{t}}(f) = \int \ell(f(x), y)\, p_{\mathrm{t}}(x)\, p_{\mathrm{s}}(y \mid x)\, dx\, dy.$$

Now multiply and divide by $p_{\mathrm{s}}(x)$:

$$R_{\mathrm{t}}(f) = \int \ell(f(x), y)\, \frac{p_{\mathrm{t}}(x)}{p_{\mathrm{s}}(x)}\, p_{\mathrm{s}}(x)\, p_{\mathrm{s}}(y \mid x)\, dx\, dy.$$

Since $p_{\mathrm{s}}(x)\, p_{\mathrm{s}}(y \mid x) = p_{\mathrm{s}}(x, y)$, we get

$$R_{\mathrm{t}}(f) = \int \ell(f(x), y)\, w(x)\, p_{\mathrm{s}}(x, y)\, dx\, dy = \mathbb{E}_{P_{\mathrm{s}}}\big[w(X)\, \ell(f(X), Y)\big],$$

where

$$w(x) = \frac{p_{\mathrm{t}}(x)}{p_{\mathrm{s}}(x)}.$$

This is the canonical covariate-shift identity. It is historically associated with Shimodaira and the later covariate-shift literature [1, 3].

**Sample version.** Given source samples $(x_i, y_i)_{i=1}^n$, a natural estimator is

$$\widehat{R}_{\mathrm{t}}(f) = \frac{1}{n} \sum_{i=1}^{n} \hat{w}(x_i)\, \ell(f(x_i), y_i),$$

where $\hat{w}$ estimates $p_{\mathrm{t}}(x)/p_{\mathrm{s}}(x)$.

**Why this works.** If the conditional label rule is stable, then source labels remain relevant; they just need to be *counted differently* because some source inputs are more representative of deployment than others.

**Algorithm: covariate-shift correction by importance weighting**

1. Train or choose a predictor $f$ on source data.
2. Estimate the density ratio $w(x) = p_{\mathrm{t}}(x)/p_{\mathrm{s}}(x)$, for example by density-ratio estimation or domain classification.
3. Form the weighted empirical risk

$$\widehat{R}_{\mathrm{t}}(f) = \frac{1}{n}\sum_{i=1}^{n} \hat{w}(x_i)\,\ell(f(x_i), y_i).$$

4. Either evaluate $f$ with this risk or retrain by minimizing the weighted loss.

**Common pitfall**

Importance weighting only helps when the covariate-shift assumption is credible and the support condition holds: if a target input region has no source data, then $p_{\mathrm{s}}(x) = 0$ and the ratio is not usable there.

### 6.3 Label shift: full derivation of posterior correction

**Derivation scaffold: label shift**

**Goal.** Correct posteriors when class priors change.
**Assumption.** $p_{\mathrm{t}}(x \mid y) = p_{\mathrm{s}}(x \mid y)$.
**Move.** Apply Bayes' rule under source and target and divide.
**Result.** $p_{\mathrm{t}}(y \mid x)$ is a reweighted version of $p_{\mathrm{s}}(y \mid x)$.

Under label shift,

$$p_{\mathrm{t}}(x \mid y) = p_{\mathrm{s}}(x \mid y), \qquad p_{\mathrm{t}}(y) \neq p_{\mathrm{s}}(y).$$

Use Bayes' rule twice:

$$p_{\mathrm{t}}(y \mid x) = \frac{p_{\mathrm{t}}(x \mid y)p_{\mathrm{t}}(y)}{p_{\mathrm{t}}(x)}, \qquad p_{\mathrm{s}}(y \mid x) = \frac{p_{\mathrm{s}}(x \mid y)p_{\mathrm{s}}(y)}{p_{\mathrm{s}}(x)}.$$

Because $p_{\mathrm{t}}(x \mid y) = p_{\mathrm{s}}(x \mid y)$, we can write

$$p_{\mathrm{t}}(y \mid x) \propto p_{\mathrm{s}}(x \mid y)\,p_{\mathrm{t}}(y).$$

Replace $p_{\mathrm{s}}(x \mid y)$ using source Bayes' rule:

$$p_{\mathrm{s}}(x \mid y) = \frac{p_{\mathrm{s}}(y \mid x)\,p_{\mathrm{s}}(x)}{p_{\mathrm{s}}(y)}.$$

Substitute:

$$p_{\mathrm{t}}(y \mid x) \propto \frac{p_{\mathrm{s}}(y \mid x)\,p_{\mathrm{s}}(x)}{p_{\mathrm{s}}(y)}\,p_{\mathrm{t}}(y).$$

For fixed $x$, the factor $p_{\mathrm{s}}(x)$ does not depend on $y$, so it is absorbed into the normalization constant.

Define the prior ratio

$$r(y) = \frac{p_{\mathrm{t}}(y)}{p_{\mathrm{s}}(y)}.$$

Then

$$p_{\mathrm{t}}(y \mid x) = \frac{r(y)\, p_{\mathrm{s}}(y \mid x)}{\sum_{y'} r(y')\, p_{\mathrm{s}}(y' \mid x)}.$$

This is the canonical label-shift correction formula [2, 15].

**What the formula says.** Take the source posterior $p_{\mathrm{s}}(y \mid x)$ from a classifier. Multiply each class by how much more common that class is at deployment. Renormalize. That is all.

**Toy example**

Suppose for a given image the source posterior is

$$p_{\mathrm{s}}(\mathrm{cat} \mid x) = 0.40, \quad p_{\mathrm{s}}(\mathrm{dog} \mid x) = 0.35, \quad p_{\mathrm{s}}(\mathrm{rabbit} \mid x) = 0.25.$$

Suppose deployment makes dogs twice as common as before, cats unchanged, rabbits half as common:

$$r(\mathrm{cat}) = 1, \quad r(\mathrm{dog}) = 2, \quad r(\mathrm{rabbit}) = 0.5.$$

The unnormalized corrected scores are

$$0.40, \quad 0.70, \quad 0.125.$$

After renormalization the posterior tilts strongly toward dog.

**Algorithm: label-shift posterior correction**

1. Train a probabilistic classifier on source data to estimate $p_{\mathrm{s}}(y \mid x)$.
2. Estimate the target prior ratios $r(y) = p_{\mathrm{t}}(y)/p_{\mathrm{s}}(y)$.
3. For each test point $x$, compute

$$\hat{p}_{\mathrm{t}}(y \mid x) = \frac{\hat{r}(y)\, \hat{p}_{\mathrm{s}}(y \mid x)}{\sum_{y'} \hat{r}(y')\, \hat{p}_{\mathrm{s}}(y' \mid x)}.$$

4. Predict using the corrected posterior.

## 6.4 From prior correction to BBSE: the confusion-matrix derivation

The remaining question is how to estimate $p_{\mathrm{t}}(y)$ or $r(y)$ without target labels. The historically important answer is *black-box shift estimation* (BBSE) [15].

Let $\hat{Y}$ be the classifier prediction. Under label shift, if the class-conditional prediction behavior stays stable, then

$$P_{\mathrm{t}}(\hat{Y} = \hat{y} \mid Y = y) = P_{\mathrm{s}}(\hat{Y} = \hat{y} \mid Y = y).$$

Define the confusion matrix $C$ by

$$C_{\hat{y},y} = P_{\mathrm{s}}(\hat{Y} = \hat{y} \mid Y = y).$$

Let $\pi_{\mathrm{t}}$ be the vector of target class priors and let $q_{\mathrm{t}}$ be the vector of target predicted-label frequencies:

$$q_{\mathrm{t}}(\hat{y}) = P_{\mathrm{t}}(\hat{Y} = \hat{y}).$$

Now marginalize over the true class $Y$:

$$q_{\mathrm{t}}(\hat{y}) = \sum_{y} P_{\mathrm{t}}(\hat{Y} = \hat{y} \mid Y = y)\, P_{\mathrm{t}}(Y = y).$$

Using the stability of the confusion matrix,

$$q_{\mathrm{t}}(\hat{y}) = \sum_{y} C_{\hat{y},y}\, \pi_{\mathrm{t}}(y).$$

In vector form,

$$q_{\mathrm{t}} = C\, \pi_{\mathrm{t}}.$$

So if $C$ is invertible,

$$\pi_{\mathrm{t}} = C^{-1} q_{\mathrm{t}}.$$

This is the core BBSE derivation. It is conceptually elegant: the target predicted-label histogram is a source confusion matrix multiplied by unknown target priors.

**Toy example**

Suppose there are two classes, cat and dog, and the source confusion matrix is

$$C = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix}.$$

The first column says that true cats are predicted as cat 90% of the time and dog 10% of the time. Assume the target predicted-label frequencies are

$$q_{\mathrm{t}} = \begin{pmatrix} 0.58 \\ 0.42 \end{pmatrix}.$$

Solving $q_{\mathrm{t}} = C\pi_{\mathrm{t}}$ recovers the target prior vector $\pi_{\mathrm{t}}$.

**Algorithm: BBSE**

1. Train a classifier on source data.
2. Estimate its source confusion matrix $C$ using held-out labeled source data.
3. Run the classifier on unlabeled target data and estimate the predicted-label frequency vector $q_{\mathrm{t}}$.

4. Solve $q_{\mathrm{t}} = C\pi_{\mathrm{t}}$ for $\pi_{\mathrm{t}}$.
5. Use $\hat{\pi}_{\mathrm{t}}$ to form prior ratios $\hat{r}(y) = \hat{\pi}_{\mathrm{t}}(y)/\hat{\pi}_{\mathrm{s}}(y)$ and apply posterior correction.

### 6.5 A historically important theoretical root: the domain-adaptation bound

**Optional advanced note**

A historically important root of the shift literature is the Ben-David et al. adaptation bound [5]. Although a novice can skip the details on first reading, the main message is conceptually important:

$$\text{target error} \leq \text{source error} + \text{domain discrepancy} + \text{best-joint-error term}.$$

In words: even a perfect source classifier may fail at deployment if the domains are far apart or if no single classifier performs well on both source and target. This result is one reason why merely reporting low source error is not enough under shift.

**What to remember**

The two classical shift identities are:

$$R_{\mathrm{t}}(f) = \mathbb{E}_{P_{\mathrm{s}}}[w(X)\ell(f(X), Y)] \quad \text{under covariate shift,}$$

and

$$p_{\mathrm{t}}(y \mid x) = \frac{r(y)p_{\mathrm{s}}(y \mid x)}{\sum_{y'} r(y')p_{\mathrm{s}}(y' \mid x)} \quad \text{under label shift.}$$

One reweights examples; the other reweights classes.

## 7 OOD detection: full derivations of the common score families

### 7.1 From hypothesis testing to score-threshold rules

**Derivation scaffold: OOD detection**

**Goal.** Decide whether $x$ came from the ID distribution or an alternative distribution.
**Ideal rule.** Compare likelihood ratio $p_{\mathrm{ID}}(x)/p_{\mathrm{OOD}}(x)$ with a threshold.
**Practical problem.** $p_{\mathrm{OOD}}$ is usually unknown and very broad.
**Result.** Use a surrogate score $S(x)$ that is higher for ID examples and threshold it.

The classical binary-testing formulation is

$$H_0 : x \sim p_{\mathrm{ID}}, \qquad H_1 : x \sim p_{\mathrm{OOD}}.$$

If both distributions were known, the Neyman–Pearson lemma says the optimal test is a threshold on the likelihood ratio

$$\Lambda(x) = \frac{p_{\mathrm{ID}}(x)}{p_{\mathrm{OOD}}(x)}.$$

But in realistic OOD detection, $p_{\mathrm{OOD}}$ is not a single known distribution. It is a huge family of possible alternatives. So most practical methods replace the ideal ratio by a *surrogate score*

$$S(x)$$

that should be large on ID data and smaller on OOD data. The detector is then

$$\phi_\tau(x) = \mathbf{1}\{S(x) \geq \tau\}.$$

This is why so many OOD papers look different but are conceptually similar: they are proposing different surrogates for the unavailable likelihood-ratio test.

### 7.2 MSP as the simplest surrogate

For a discriminative classifier with logits $z_y(x)$ and softmax posterior $p_\theta(y \mid x)$, the MSP score is

$$S_{\mathrm{MSP}}(x) = \max_y p_\theta(y \mid x).$$

It is not derived from a full generative model. Instead it uses a practical heuristic:

> *If the classifier strongly prefers one training class, treat the example as more likely to be ID.*

This heuristic is historically important because it is simple, cheap, and still strong [11]. It also makes clear what later methods try to improve: MSP only sees relative class probabilities and can be overconfident on unfamiliar inputs.

### 7.3 Energy scores from log-sum-exp

Energy-based OOD detection uses

$$S_{\mathrm{energy}}(x) = T \log \sum_{y=1}^{K} \exp(z_y(x)/T).$$

To see where this comes from, recall the log-partition function

$$A(z) = \log \sum_{y=1}^{K} e^{z_y}.$$

The softmax is the gradient of this log-partition:

$$\frac{\partial A(z)}{\partial z_y} = \frac{e^{z_y}}{\sum_k e^{z_k}} = \mathrm{softmax}_y(z).$$

So $A(z)$ is a scalar summary of the entire logit vector, not only the largest relative probability. Multiplying by $T$ gives a temperature-controlled version.

A useful algebraic identity is

$$\max_y z_y \leq \log \sum_y e^{z_y} \leq \max_y z_y + \log K.$$

So energy tracks the overall logit level and is closely related to the maximum logit while still aggregating information across classes. Liu et al. showed that this can separate ID and OOD better than MSP in many settings [23].

**Why MSP and energy differ.** Suppose we add a constant $c$ to all logits. Softmax probabilities do not change, so MSP is unchanged. But

$$T \log \sum_y e^{(z_y+c)/T} = c + T \log \sum_y e^{z_y/T}.$$

Thus energy shifts by $c$. This is the formal reason energy can use absolute logit scale while MSP cannot.

## 7.4 Mahalanobis scores from a Gaussian feature model

Mahalanobis OOD detection [14] is often presented as a feature-space heuristic, but it has a clean derivation from a Gaussian discriminant model.

Assume the feature representation $\phi(x) \in \mathbb{R}^d$ satisfies

$$\phi(x) \mid Y = y \sim \mathcal{N}(\mu_y, \Sigma),$$

with class-specific means $\mu_y$ and a shared covariance $\Sigma$. Then the class-conditional log-density is

$$\log p(\phi(x) \mid y) = -\frac{1}{2}(\phi(x) - \mu_y)^\top \Sigma^{-1}(\phi(x) - \mu_y) - \frac{1}{2}\log|\Sigma| - \frac{d}{2}\log(2\pi).$$

The last two terms do not depend on $y$, so comparing classes by log-density is equivalent to comparing

$$-\frac{1}{2}(\phi(x) - \mu_y)^\top \Sigma^{-1}(\phi(x) - \mu_y).$$

Therefore the most ID-like class under this model is the one with smallest Mahalanobis distance. This motivates the score

$$S_{\text{Mah}}(x) = -\min_y (\phi(x) - \mu_y)^\top \Sigma^{-1}(\phi(x) - \mu_y).$$

**Interpretation.** The shared covariance rescales directions according to how much variability is expected inside ID data. Distances in unusually stable directions matter more.

### 7.5 ODIN: why temperature scaling and tiny input perturbations help

ODIN [13] modifies the score before thresholding in two ways.

First, it applies a high temperature $T$ in the softmax:

$$p_T(y \mid x) = \frac{e^{z_y(x)/T}}{\sum_k e^{z_k(x)/T}}.$$

Second, it perturbs the input in the direction that *increases* the confidence of the predicted class. If $\hat{y}(x) = \arg\max_y p_T(y \mid x)$, ODIN uses

$$x' = x - \varepsilon \ \mathrm{sign}\Big( -\nabla_x \log p_T(\hat{y}(x) \mid x)\Big).$$

Equivalently, it adds a small perturbation that increases $\log p_T(\hat{y}(x) \mid x)$.

Why can this help? The local confidence landscape around ID inputs can respond differently to this perturbation than the landscape around OOD inputs. In practice, ODIN often sharpens separation even though it still ends with a threshold on a confidence score.

### 7.6 OpenMax and Outlier Exposure as historical bridges

OpenMax [9] is historically important because it explicitly tries to reserve probability mass for an "unknown" class. It models class activation distances and calibrates tail behavior with extreme-value modeling. Conceptually, it says:

*Do not force all probability mass onto known classes if the activation pattern looks far from every known class prototype.*

Outlier Exposure (OE) [18] makes a different move. Instead of changing the test-time score alone, it changes training. Let $\mathcal{D}_{\text{in}}$ be ID data and $\mathcal{D}_{\text{out}}^{\text{aux}}$ be auxiliary outliers. A standard OE objective is

$$\mathcal{L}_{\text{OE}} = \underbrace{\mathbb{E}_{(x,y)\sim\mathcal{D}_{\text{in}}}\big[-\log p_\theta(y \mid x)\big]}_{\text{ordinary classification loss}} + \lambda \underbrace{\mathbb{E}_{x\sim\mathcal{D}_{\text{out}}^{\text{aux}}}\big[\mathcal{H}(u, p_\theta(\cdot \mid x))\big]}_{\text{encourage low-confidence / near-uniform output on auxiliary outliers}},$$

where $u$ is the uniform distribution over classes and $\mathcal{H}$ is cross-entropy.

OE still produces a score-threshold detector at test time, but it changes the geometry of the classifier so that many non-ID inputs receive lower confidence.

### 7.7 Thresholds, metrics, and calibration conventions

Three reporting conventions are standard.

- **AUROC:** threshold-free ranking quality between ID and OOD scores.
- **AUPR:** precision-recall area, useful when OOD is rare.

- **FPR95:** false-positive rate on OOD when ID true-positive rate is fixed at 95%.

A beginner should understand FPR95 carefully. If we tune $\tau$ so that 95% of ID examples are accepted, FPR95 is the fraction of OOD examples that still slip through. Lower is better.

**Common pitfall**

A paper can improve AUROC while still being unusable in deployment if the thresholded behavior at the desired operating point is poor. That is why threshold-free and thresholded metrics should both be reported.

**What to remember**

The common OOD families can be read as increasingly structured approximations to the unavailable ideal ratio test:

$$\text{MSP} \to \text{energy} \to \text{feature-density / Mahalanobis} \to \text{training with OE}.$$

All of them end in the same final action: compute a scalar score and compare it with a threshold.

# 8 Adversarial robustness: from min–max risk to attacks and defenses

## 8.1 Robust risk and the min–max objective

**Derivation scaffold: adversarial robustness**

**Goal.** Measure and train performance under worst-case allowed perturbations.
**Assumption.** A threat model specifies the perturbation set $\Delta(x)$, often $\{\delta : \|\delta\|_p \le \varepsilon\}$.
**Move.** Replace ordinary loss by the maximum loss inside that set.
**Result.** Robust learning becomes min–max optimization.

For each clean example $(x, y)$, define the allowed perturbation set

$$\Delta(x) = \{\delta : \|\delta\|_p \le \varepsilon,\ x + \delta \in [0,1]^d\}.$$

The robust loss at $(x, y)$ is

$$\ell_{\text{rob}}(f; x, y) = \max_{\delta \in \Delta(x)} \ell(f(x+\delta), y).$$

The robust risk is then

$$R_{\text{rob}}(f) = \mathbb{E}_{(X,Y)}\Big[\max_{\delta \in \Delta(X)} \ell(f(X+\delta), Y)\Big].$$

Training for adversarial robustness means solving

$$\min_{\theta} \mathbb{E}_{(X,Y)}\Big[\max_{\delta \in \Delta(X)} \ell(f_\theta(X+\delta), Y)\Big].$$

This is the canonical min–max formulation emphasized by Madry et al. [51].

### 8.2 FGSM: full derivation in the $\ell_\infty$ case

Let
$$g = \nabla_x \ell(f_\theta(x), y).$$
For a small perturbation $\delta$, first-order Taylor expansion gives
$$\ell(f_\theta(x+\delta), y) \approx \ell(f_\theta(x), y) + g^\top \delta.$$
Under the $\ell_\infty$ constraint $|\delta_i| \le \varepsilon$ for each coordinate, we need to maximize
$$g^\top \delta = \sum_{i=1}^{d} g_i \delta_i.$$
Each coordinate can be optimized independently:
$$\max_{|\delta_i| \le \varepsilon} g_i \delta_i = \varepsilon |g_i|,$$
achieved by
$$\delta_i^\star = \varepsilon \ \mathrm{sign}(g_i).$$
Therefore
$$\delta^\star = \varepsilon \ \mathrm{sign}(g) = \varepsilon \ \mathrm{sign}(\nabla_x \ell(f_\theta(x), y)).$$
This is FGSM.

**Why the sign appears.** It is not a magic heuristic. It is the exact optimizer of the linearized inner problem under an $\ell_\infty$ constraint.

### 8.3 General dual-norm derivation

The $\ell_\infty$ case is the easiest for beginners, but the general pattern is also important. Consider
$$\max_{\|\delta\|_p \le \varepsilon} g^\top \delta.$$
By Hölder's inequality,
$$g^\top \delta \le \|g\|_q \|\delta\|_p \le \varepsilon \|g\|_q,$$
where $q$ is the dual norm satisfying $1/p + 1/q = 1$. Thus the optimal value is $\varepsilon\|g\|_q$, and the optimizer aligns with the dual-norm direction. This shows that FGSM is the special $p = \infty,\ q = 1$ case of a broader first-order principle.

### 8.4 PGD and adversarial training

The inner maximization is usually nonconvex for deep networks, so we solve it approximately. PGD iterates

$$\delta^{t+1} = \Pi_{\|\delta\|_p \le \varepsilon}(\delta^t + \alpha\, u^t),$$

where $u^t$ is a gradient-ascent direction such as $\mathrm{sign}(\nabla_\delta \ell)$ for $\ell_\infty$ attacks. The approximate maximizer $\hat{\delta}(x)$ is then used inside training:

$$\min_\theta \frac{1}{n} \sum_{i=1}^{n} \ell(f_\theta(x_i + \hat{\delta}(x_i)), y_i).$$

**Algorithm: PGD adversarial training**

For each minibatch:

1. Initialize $\delta^0$ inside the perturbation ball, often randomly.
2. Run $T$ PGD ascent steps to approximately maximize loss.
3. Form adversarial examples $x_i + \delta_i^T$.
4. Update $\theta$ by descending the loss on these adversarial examples.

This is computationally expensive because every training step contains an inner attack loop. But it remains the canonical empirical defense baseline.

### 8.5 TRADES: derivation of the standard robustness–accuracy objective

TRADES [21] starts from a decomposition of robust error into natural error plus a boundary term. The exact theorem uses a classification-calibrated surrogate argument; the core idea is easier to understand in plain language:

*A point can be correctly classified on the clean input and still be dangerously close to the decision boundary.*

So TRADES keeps the ordinary classification loss on the clean example, but adds a term that penalizes prediction change inside the perturbation ball:

$$\mathcal{L}_{\mathrm{TRADES}}(x, y) = \underbrace{-\log p_\theta(y \mid x)}_{\text{natural accuracy term}} + \beta \underbrace{\max_{\|\delta\|_p \le \varepsilon} \mathrm{KL}\left(p_\theta(\cdot \mid x) \,\|\, p_\theta(\cdot \mid x + \delta)\right)}_{\text{boundary-stability term}}.$$

The inner maximization finds a perturbation that changes the predictive distribution as much as possible. The outer minimization makes the predictive distribution stable inside the perturbation set. The scalar $\beta$ controls the trade-off between clean accuracy and robustness.

**Why KL divergence appears.** KL divergence measures how much the predicted class distribution changes under perturbation. If we want the boundary to be locally stable, it is natural to penalize this change directly.

**Toy example**

Imagine a clean image sits just barely on the dog side of the cat–dog boundary. Cross-entropy on the clean point alone may be small, so ordinary training is satisfied. But a tiny perturbation can push the point across the boundary. TRADES detects this by maximizing the KL divergence between the clean predictive distribution and the perturbed predictive distribution.

## 8.6 A canonical certified example: interval bound propagation (IBP)

Empirical robustness uses attacks to *search* for adversarial examples. Certified robustness tries to *prove* that none exist inside the perturbation set.

A beginner-friendly certified method is interval bound propagation (IBP) [17]. Suppose the input interval is

$$x \in [\underline{x}, \overline{x}] = [x - \varepsilon,\ x + \varepsilon]$$

coordinatewise under an $\ell_\infty$ budget. For an affine layer

$$h = Wx + b,$$

the interval bounds propagate as

$$\underline{h} = W^+\underline{x} + W^-\overline{x} + b, \qquad \overline{h} = W^+\overline{x} + W^-\underline{x} + b,$$

where $W^+_{ij} = \max(W_{ij}, 0)$ and $W^-_{ij} = \min(W_{ij}, 0)$. For a monotone nonlinearity such as ReLU,

$$\underline{z} = \mathrm{ReLU}(\underline{h}), \qquad \overline{z} = \mathrm{ReLU}(\overline{h}).$$

Propagating these intervals layer by layer gives bounds on every output logit. If for the true class $y$,

$$\underline{z}_y > \overline{z}_k \qquad \text{for all } k \neq y,$$

then the classifier's prediction cannot change anywhere inside the perturbation box. That is a certificate.

**Why this matters conceptually.** Attacks show *lower bounds* on vulnerability: "I found an adversarial example." Certification aims for *upper bounds*: "I can prove none exist in this region." The two play different scientific roles.

## 8.7 Evaluation and reporting conventions

A careful robustness report should include:

- clean accuracy;
- robust accuracy under the stated threat model;
- exact attack hyperparameters;
- whether multi-restart attacks or AutoAttack were used;
- whether the defense still works under distribution shift or only on the benchmark distribution [44].

**What to remember**

The canonical adversarial framework is

$$\min_\theta \mathbb{E}\Big[\max_{\delta\in\Delta(X)} \ell(f_\theta(X+\delta), Y)\Big].$$

FGSM solves a one-step linearized inner problem, PGD approximates the full inner maximization, TRADES adds an explicit clean/robust trade-off term, and certification tries to prove the absence of adversarial examples rather than merely failing to find them.

# 9 Conformal prediction under shift: from rank arguments to weighted and adaptive methods

## 9.1 Split conformal prediction: exact coverage from rank symmetry

**Derivation scaffold: split conformal**

**Goal.** Build a prediction set with finite-sample marginal coverage $1-\alpha$.
**Assumption.** Calibration examples and the test point are exchangeable.
**Move.** Use the rank of the test score among calibration scores plus the test score.
**Result.** Threshold at an empirical quantile.

Let $(X_i, Y_i)_{i=1}^n$ be calibration examples and let $V(X_i, Y_i)$ be their nonconformity scores, where larger values mean "less compatible." For a test input $x$ and candidate label $y$, compute $V(x, y)$.

Define the split-conformal prediction set

$$C(x) = \{y : V(x, y) \le \hat{q}\},$$

where $\hat{q}$ is the

$$k = \lceil (n+1)(1-\alpha) \rceil$$

th smallest calibration score.

Why does this work? Let $V_{n+1} = V(X_{n+1}, Y_{n+1})$ be the score of the future test example with its true label. Under exchangeability, the multiset

$$V_1, \dots, V_n, V_{n+1}$$

is exchangeable, so the rank of $V_{n+1}$ among these $n+1$ values is uniform over $\{1,\ldots,n+1\}$ up to ties. Therefore

$$\mathbb{P}\{V_{n+1} > \hat{q}\} \le \alpha.$$

Equivalently,

$$\mathbb{P}\{Y_{n+1} \in C(X_{n+1})\} \ge 1-\alpha.$$

This is the finite-sample split-conformal guarantee [10, 12].

### 9.2 APS and RAPS: detailed classification-set formulas

For classification, APS and RAPS use score functions tailored to ranked class probabilities.

Let $\hat{p}_{(1)}(x) \ge \cdots \ge \hat{p}_{(K)}(x)$ be sorted predicted probabilities, and let $r_y(x)$ be the rank of label $y$. The APS score is

$$V_{\mathrm{APS}}(x,y) = \sum_{j=1}^{r_y(x)} \hat{p}_{(j)}(x).$$

This score is small if $y$ appears early in the ranking and large if $y$ appears only after accumulating lots of probability mass.

RAPS adds a regularization penalty for deep ranks:

$$V_{\mathrm{RAPS}}(x,y) = \sum_{j=1}^{r_y(x)} \hat{p}_{(j)}(x) + \lambda\,(r_y(x) - k_{\mathrm{reg}})_+.$$

Once a label falls below rank $k_{\mathrm{reg}}$, the extra penalty discourages unnecessarily large sets. Romano et al. and Angelopoulos et al. developed these classification-set constructions and showed they are both valid and practically efficient [22, 25].

### 9.3 Weighted conformal under covariate shift

Ordinary split conformal treats every calibration example equally. Under covariate shift, some calibration points are much more representative of the target than others. Weighted conformal prediction addresses this [19].

Let $w(x) = p_{\mathrm{t}}(x)/p_{\mathrm{s}}(x)$. For a test point $x$, assign calibration weights

$$\tilde{w}_i(x) = w(X_i), \qquad \tilde{w}_{n+1}(x) = w(x).$$

Normalize:

$$p_i^w(x) = \frac{\tilde{w}_i(x)}{\sum_{j=1}^{n+1} \tilde{w}_j(x)}, \qquad i = 1,\ldots,n+1.$$

Now form a weighted empirical distribution on the calibration scores, plus an atom at $+\infty$ carrying the test-point weight. Let $V_i = V(X_i, Y_i)$. Define the weighted threshold as the smallest $t$ such that

$$\sum_{i=1}^{n} p_i^w(x)\, \mathbf{1}\{V_i \le t\} \ge 1 - \alpha.$$

Call this threshold $\hat{q}_w(x)$. Then output

$$C_w(x) = \{y : V(x, y) \le \hat{q}_w(x)\}.$$

**Why the extra mass at the test point appears.** Ordinary conformal effectively compares the test score against $n$ calibration scores *plus itself*. Under weighting, the test point should also receive weight. The "atom at infinity" is the weighted way to reserve room for that unseen test score.

## 9.4 Label-shift conformal prediction

Under label shift, class priors change while $p(x \mid y)$ remains stable. A simple conformal response is to reweight calibration examples by a class-dependent ratio

$$r(y) = \frac{p_{\mathrm{t}}(y)}{p_{\mathrm{s}}(y)}.$$

If $V_i = V(X_i, Y_i)$ is the calibration score on example $i$, assign weight

$$\omega_i = r(Y_i).$$

Then use a weighted quantile of the calibration scores with these label-dependent weights. Intuitively, calibration examples from classes that are more common at deployment should count more heavily. Podkopaev and Ramdas formalized distribution-free uncertainty quantification for classification under label shift in this spirit [27].

## 9.5 Adaptive conformal inference (ACI) for drifting streams

When data arrive sequentially and the distribution drifts over time, even weighted offline calibration may be too static. ACI [26] updates the target miscoverage level online.

A standard ACI update is

$$\alpha_{t+1} = \alpha_t + \eta(\alpha - \mathrm{err}_t),$$

where $\mathrm{err}_t = \mathbf{1}\{Y_t \notin C_t(X_t)\}$ is the observed miscoverage at time $t$, and $\eta > 0$ is a step size. The next prediction set is built using $\alpha_{t+1}$ rather than a fixed $\alpha$.

The intuition is simple:

- if we are missing too often ($\mathrm{err}_t = 1$ frequently), shrink $\alpha_t$ so the next sets become larger;
- if we are too conservative, increase $\alpha_t$ so the next sets become smaller.

### 9.6 How later online variants modify ACI

The online conformal literature after ACI is easier to understand once ACI is seen as an online control rule.

- **DtACI / online arbitrary-shift methods** modify the update to react faster and achieve local guarantees over drifting windows [34].
- **Conformal PID control** adds integral and derivative style correction terms, borrowing language from control theory, to track systematic drift more stably [31].
- **Error-quantified conformal inference (ECI)** uses not only whether a miss occurred, but by how much the current score exceeded the threshold, thereby replacing binary feedback with smoother error feedback [50].

A novice does not need every proof at first pass. The main point is that these methods preserve the same skeleton:

$$\text{score} \to \text{threshold} \to \text{online threshold update.}$$

### 9.7 A note on unlabeled adaptation and multi-source conformal

Two recent directions deserve special mention.

First, unlabeled target adaptation methods such as ECP/EACP adjust conformal scores using the model's uncertainty on unlabeled target data [35]. The goal is not to recover exact exchangeability, but to produce smaller or better-shaped sets under shift when labels are unavailable.

Second, multi-source conformal inference uses several source domains and a target domain. The statistical challenge is to borrow strength without trusting biased sources equally. Recent work derives efficient influence-function based target quantiles and source-reweighting rules [36]. For a beginner, the high-level message is enough: instead of one calibration pool, there are many, and they are weighted according to relevance to the target.

**What to remember**

Conformal prediction always reduces uncertainty quantification to a calibrated threshold on a score. Under no shift, exchangeability gives the threshold. Under covariate shift, weights change which calibration examples matter. Under label shift, class-dependent weights matter. Under online drift, the threshold itself must adapt over time.

## Part III

# Part III: Historical map and recent influential works

# 10 Historical map: roots to leaves

This section is not meant to overwhelm a novice with dozens of papers. Its purpose is to show how the field grew. A good reading rule is:

*Read one root paper, one canonical paper, and one recent benchmark-shaping paper in each topic.*

## 10.1 A compact historical timeline

| **Year** | Work | Framework connection | Why it mattered historically |
|---|---|---|---|
| **2000** | Shimodaira [1] | Covariate-shift risk reweighting | Classical importance-weighting identity for target risk under covariate shift. |
| **2002** | Saerens et al. [2] | Label-shift posterior correction | Early and influential prior-correction view. |
| **2005** | Vovk, Gammerman, and Shafer [10] | Conformal prediction | Foundational exchangeability-and-rank view behind prediction sets with finite-sample validity. |
| **2007** | Sugiyama et al. [3] | Covariate-shift estimation | Consolidated methods and theory for importance weighting and density-ratio estimation. |
| **2009** | Quionero-Candela et al. [4] | Dataset-shift taxonomy | Standard reference for dataset-shift language and cases. |
| **2010** | Ben-David et al. [5] | Domain adaptation theory | Showed that target error depends on source error, domain discrepancy, and shared difficulty. |
| **2012** | Moreno-Torres et al. [6] | Dataset-shift taxonomy | Clarified terminology and practical distinctions among shift types. |
| **2014** | Szegedy et al. [7] | Adversarial examples | Brought adversarial vulnerability into mainstream deep-learning discussion. |
| **2015** | Goodfellow et al. [8] | FGSM / first-order attacks | Gave the linearization view that still anchors beginner understanding. |
| **2016** | Bendale and Boult [9] | Open-set / OOD | Introduced OpenMax, a major historical bridge from open-set recognition to modern OOD detection. |
| **2017** | Hendrycks and Gimpel [11] | MSP baseline | Established maximum softmax probability as the canonical OOD baseline. |
| **2018** | Liang et al. [13] | OOD score sharpening | ODIN showed that temperature scaling and input preprocessing improve confidence-based detection. |
| **2018** | Lee et al. [14] | Feature-density OOD | Popularized Mahalanobis feature-space scoring. |
| **2018** | Lipton et al. [15] | BBSE | Confusion-matrix inversion for estimating target class priors without target labels. |
| **2018** | Madry et al. [51] | Min–max robustness | Set the standard adversarial-training objective and PGD-based evaluation philosophy. |
| **2018** | Athalye et al. [16] | Robustness evaluation | Exposed gradient masking and evaluation failures. |
| **2018** | Gowal et al. [17] | Certified robustness | IBP became a canonical beginner entry point to certification. |
| **2019** | Hendrycks et al. [18] | OOD training with negatives | Outlier Exposure taught detectors to say “no” using auxiliary outliers. |

| **Year** | Work | Framework connection | Why it mattered historically |
|---|---|---|---|
| **2019** | Tibshirani et al. [19] | Covariate-shift conformal | Weighted conformal prediction under covariate shift. |
| **2019** | Tsipras et al. [20] | Robustness/accuracy tension | Made the trade-off discussion central. |
| **2019** | Zhang et al. [21] | TRADES | Canonical objective for balancing clean accuracy and robustness. |
| **2020** | Romano et al. [22] | APS / adaptive sets | Strong classification-set construction for conformal prediction. |
| **2020** | Liu et al. [23] | Energy OOD | Energy score became a central post-hoc OOD family. |
| **2020** | Croce and Hein [24] | Robustness evaluation | AutoAttack set a higher standard for attack-based evaluation. |
| **2021** | Angelopoulos et al. [25] | RAPS | Refined conformal classification sets to improve efficiency. |
| **2021** | Gibbs and Candès [26] | Adaptive conformal | Online threshold adaptation under drift. |
| **2021** | Podkopaev and Ramdas [27] | Label-shift conformal | Distribution-free uncertainty under class-prior shift. |
| **2021** | Yang et al. [28] | OOD/open-set survey | Useful bridge survey for the pre-foundation-model era. |
| **2023** | Tao et al. [29] | OOD with synthesized negatives | NPOS pushed near-boundary synthetic outliers. |
| **2023** | Angelopoulos and Bates [30] | Conformal pedagogy | Organized a fast-growing field into a common language. |
| **2023** | Angelopoulos et al. [31] | Online conformal control | PID-style threshold updates for nonstationary time series. |
| **2023** | Bitterwolf et al. [32] | OOD benchmark repair | Showed that benchmark contamination can distort conclusions. |
| **2024** | Du et al. [33] | OOD from wild unlabeled data | SAL filtered useful proxy negatives from contaminated unlabeled data. |
| **2024** | Gibbs and Candès [34] | Online conformal theory | Sharper local-regret style guarantees under arbitrary shift. |
| **2024** | Kasa et al. [35] | Unlabeled target conformal | Adapted conformal scores using unlabeled target uncertainty. |
| **2024** | Liu et al. [36] | Multi-source conformal | Target quantiles estimated from multiple domains. |
| **2024** | Xu et al. [37] | Joint-shift conformal robustness | Shaped score distributions across multiple domains. |
| **2024** | Yang et al. [38] | OOD benchmark semantics | Cleaned the semantic-shift question at ImageNet scale. |
| **2024** | Zhang et al. [39] | Standardized OOD evaluation | OpenOOD v1.5 widened and normalized protocol choices. |
| **2024** | Long et al. [40] | Graded OOD evaluation | Argued that binary in/out views can hide graded shift structure. |
| **2024** | Li et al. [44] | Robustness under shift benchmarks | Showed that robustness claims can collapse under dataset-wise or threat-wise shift. |
| **2024** | Wang et al. [45] | Scaled adversarial training | Brought min–max training into the foundation-model regime. |

| Year | Work | Framework connection | Why it mattered historically |
|---|---|---|---|
| **2024** | Schlarmann et al. [46] | VLM robustness | Robustified CLIP encoders for downstream vision–language models. |
| **2024** | Angelopoulos, Barber, and Bates [48] | Conformal theory synthesis | Unified proof patterns and pedagogy for the next generation of work. |
| **2025** | Noda et al. [49] | Real-world VLM OOD evaluation | Pushed the field toward harder semantic and full-spectrum benchmarks with VLMs. |
| **2025** | Wu et al. [50] | Online conformal refinement | Continuous error feedback instead of binary hit/miss updates. |

### 10.2 How to read the historical map

A novice can compress the whole map into four "trunks."

- **Distribution shift trunk:** target-risk correction by reweighting or prior correction → theoretical adaptation bounds → uncertainty quantification under shift.
- **OOD trunk:** open-set recognition → confidence baselines → better score families → training with auxiliary outliers → benchmark correction and full-spectrum evaluation.
- **Adversarial trunk:** discovery of adversarial examples → first-order attacks → min–max training → trade-off objectives → stronger evaluation → scaling and multimodal robustness.
- **Conformal trunk:** exchangeability and rank arguments → classification-specific scores → shift-aware weighting → online adaptation → multi-source and unlabeled-target extensions.

**What to remember**

The point of historical reading is not memorizing years. It is recognizing that most modern work changes one of only a few ingredients:

- which distributional assumption is made;
- which score is used;
- which data are used for calibration;
- which benchmark makes the claim believable.

## 11 Recent influential OOD works (2023–2025)

This section keeps a fixed template for each work:

relation to canonical framework →
what changed →
what it claimed →
how the claim was supported.

## 11.1 Papers that most changed community practice

**Recent-work note: NINCO: fix the benchmark before comparing detectors**

**Relation to the canonical framework.** NINCO changes no detector score and no training objective. It changes the *evaluation distribution*.
**Key idea.** If the nominal OOD benchmark $Q$ is actually a mixture

$$Q = (1-\eta)Q_{\text{true-ood}} + \eta Q_{\text{id-cont}},$$

then the measured false-positive rate becomes

$$\mathrm{FPR}_Q(\tau) = (1-\eta)\mathrm{FPR}_{Q_{\text{true-ood}}}(\tau) + \eta \mathbb{P}_{x\sim Q_{\text{id-cont}}}(S(x) \geq \tau).$$

A sensible detector can therefore be penalized for assigning high ID scores to contaminated examples.
**Claim.** Many ImageNet OOD comparisons were distorted by ID contamination.
**Evidence.** Careful dataset curation and empirical reevaluation on the cleaned NINCO benchmark [32].

**Recent-work note: ImageNet-OOD: separate semantic novelty from covariate artifacts**

**Relation to the canonical framework.** Still score-threshold OOD detection. The change is *what success means*.
**Key idea.** Benchmark construction should minimize covariate mismatch so that performance reflects semantic novelty rather than spurious visual differences.
**Claim.** Modern detectors often look strong because they respond to covariate shift more than to semantic shift.
**Evidence.** A curated ImageNet-scale benchmark and broad empirical comparison of contemporary detectors [38].

**Recent-work note: OpenOOD v1.5 and the move toward standardized full-spectrum evaluation**

**Relation to the canonical framework.** Same detector interface, much better protocol.
**What changed.** OpenOOD v1.5 broadened evaluation across near-OOD, far-OOD, covariate shift, ImageNet-scale settings, and foundation-model backbones [39].
**Claim.** Detector gains should survive standardized, multi-setting evaluation rather than one favorable dataset pair.
**Evidence.** Unified benchmark tooling and large-scale reevaluation.

**Recent-work note: OOD-X / real-world VLM benchmarks (2025)**

**Relation to the canonical framework.** Same score-threshold view, but now the representation often comes from a VLM or CLIP-like encoder.
**Key formulas.** For a VLM with image embedding $g(x)$ and text embeddings $e(t_k)$,

$$a_k(x) = \frac{\langle g(x), e(t_k)\rangle}{T}, \qquad S_{\text{energy-VLM}}(x) = T \log \sum_k e^{a_k(x)}.$$

So the score family is familiar even though the representation changed.
**Claim.** Older OOD benchmarks are saturated; modern CLIP-based methods need harder semantic and full-spectrum tests.
**Evidence.** New benchmarks such as ImageNet-X, ImageNet-FS-X, and Wilds-FS-X and extensive reevaluation of VLM-based detectors [49, 47].

## 11.2 Algorithmic recent works

**Recent-work note: NPOS**

**Relation to the canonical framework.** Still score-threshold OOD detection, but the score is trained using synthesized near-boundary negatives rather than only post-hoc confidence.
**Key idea.** Use representation-space geometry to synthesize proxy outliers near the ID boundary. A simplified training objective is

$$\min_{\theta} \mathbb{E}_{\mathrm{ID}} \ell_{\mathrm{cls}} + \lambda \Big( \mathbb{E}_{z \sim \mathrm{ID}} \log \sigma(u_\theta(z)) + \mathbb{E}_{z' \sim \mathrm{syn}} \log(1 - \sigma(u_\theta(z'))) \Big).$$

**Claim.** Non-parametric outlier synthesis produces more useful near-OOD negatives than simple Gaussian or heuristic generation.
**Evidence.** Strong empirical gains across standard OOD benchmarks [29].

**Recent-work note: SAL**

**Relation to the canonical framework.** SAL still ends with a learned OOD score, but it first mines proxy negatives from contaminated unlabeled wild data.
**Key idea.** Filter wild samples using a gradient-geometry score

$$\tau_i = \Big\langle \nabla \ell(h(\tilde{x}_i), \hat{y}_i) - \bar{g},\ v \Big\rangle^2,$$

select likely outliers $\mathcal{S}_T = \{\tilde{x}_i : \tau_i > T\}$, then train a binary OOD classifier using labeled ID data and $\mathcal{S}_T$.
**Claim.** Unlabeled wild data are useful if we can first separate likely outliers from likely ID points.
**Evidence.** Theoretical filtering guarantees and strong empirical results [33].

## 11.3 Why these OOD works mattered broadly

The big recent OOD story is not merely "a better score beat an older score." The broader story is:

1. benchmark repair and standardization changed which claims are believable;
2. semantic shift was disentangled from covariate artifacts;
3. foundation models and VLMs changed the representation layer while keeping the score-threshold skeleton;
4. no single method dominates once evaluation becomes realistic.

That is why the benchmark papers are at least as influential as the algorithmic ones.

# 12 Recent influential adversarial-robustness works (2023–2025)

This section explicitly distinguishes **community-shaping benchmark or paradigm papers** from **more specialized algorithmic refinements**. That distinction matters for students, because not every technically good paper has the same field-wide impact.

## 12.1 Community-shaping recent works

**Recent-work note: OODRobustBench**

**Relation to the canonical framework.** Replace in-distribution robust risk

$$R_{\mathrm{rob,s}}(\theta) = \mathbb{E}_{P_{\mathrm{s}}}\left[\max_{\delta\in\Delta_{\mathrm{s}}} \ell(f_\theta(X+\delta), Y)\right]$$

by the more deployment-relevant object

$$R_{\mathrm{rob,t}}(\theta) = \mathbb{E}_{P_{\mathrm{t}}}\left[\max_{\delta\in\Delta_{\mathrm{t}}} \ell(f_\theta(X+\delta), Y)\right].$$

**Claim.** Robustness on the training benchmark does not by itself characterize robustness under dataset-wise or threat-wise shift.
**Evidence.** Large-scale evaluation of 706 robust models across 23 dataset-wise shifts and 6 threat-wise shifts [44].
**Why it mattered.** It changed how robustness results should be interpreted in deployment-oriented work.

**Recent-work note: AdvXL**

**Relation to the canonical framework.** Same Madry min–max objective, different optimization schedule.
**Key formulas.** Stage 1 adversarial pretraining solves a cheaper objective

$$\theta^{(1)} = \arg\min_\theta \mathbb{E}\left[\max_{\delta\in\Delta_{\mathrm{weak}}} \ell(f_\theta(r(x)+\delta), y)\right],$$

then stage 2 full-resolution fine-tuning solves

$$\theta^{(2)} = \arg\min_\theta \mathbb{E}\left[\max_{\delta\in\Delta_{\mathrm{strong}}} \ell(f_\theta(x+\delta), y)\right].$$

**Claim.** Adversarial training can benefit from foundation-model scale if compute bottlenecks are handled properly.
**Evidence.** New strong AutoAttack results on ImageNet-1K and gains under multiple norms [45].
**Why it mattered.** It re-opened adversarial training in the large-model, web-scale-data era.

**Recent-work note: Robust CLIP**

**Relation to the canonical framework.** Still min–max optimization, but the outer loss is representation preservation rather than supervised cross-entropy.
**Key objective.**

$$L_{\mathrm{FARE}}(\phi, x) = \max_{\|z-x\|_\infty\le\varepsilon} \|\phi(z) - \phi_{\mathrm{Org}}(x)\|_2^2.$$

**Claim.** Robustifying the CLIP image encoder transfers robustness to many downstream VLM systems without retraining them.
**Evidence.** Gains in adversarial robustness across zero-shot classification and large vision-language-model tasks [46].
**Why it mattered.** It connected adversarial robustness to the VLM ecosystem rather than only supervised image classifiers.

## 12.2 Algorithmic refinements that are still useful to know

**Recent-work note: Subset adversarial training**

**Relation to the canonical framework.** Sparsifies the inner maximization over examples instead of attacking every example at every step.

**Key objective.**

$$\min_{\theta}\left[\frac{1}{|A|}\sum_{(x,y)\in A}\max_{\delta\in\Delta}\ell(f_\theta(x+\delta),y)+\frac{1}{|B|}\sum_{(x,y)\in B}\ell(f_\theta(x),y)\right].$$

**Claim.** Robustness transfer is broader than a naive per-example interpretation suggests, so compute can be reduced.
**Evidence.** Empirical study across subsets, classes, and tasks [41].

**Recent-work note: RAMP**

**Relation to the canonical framework.** Expands the threat model from one norm to a union of norms.
**Key idea.** Train against

$$\Delta_{\text{union}} = B_1(\varepsilon_1)\cup B_2(\varepsilon_2)\cup B_\infty(\varepsilon_\infty)$$

and add multi-norm coupling terms so clean and robust performance do not collapse.
**Claim.** A single defense can be robust across several perturbation geometries.
**Evidence.** Strong union-robustness results and good clean accuracy [42].

**Recent-work note: Conflict-Aware Adversarial Training**

**Relation to the canonical framework.** Leaves the adversarial inner problem intact but changes how clean and adversarial gradients are reconciled.
**Key idea.** If

$$g_{\text{std}} = \nabla_\theta L_{\text{std}}, \qquad g_{\text{adv}} = \nabla_\theta L_{\text{adv}},$$

then project or modify the descent direction when these gradients conflict too strongly.
**Claim.** The clean/robust trade-off is partly a gradient-conflict problem, not only a scalar weighting problem.
**Evidence.** Empirical gains over fixed-weight baselines under multiple attack budgets [43].

### 12.3 Why these adversarial works mattered broadly

The broad recent adversarial story is:

1. evaluation moved from "robust on the benchmark" toward "robust under shift too";
2. adversarial training entered the scaling era;
3. robustness work started to interact directly with CLIP/VLM systems;
4. several 2024 papers refined the optimization details of the canonical min–max framework rather than replacing it.

For a student, the main lesson is that the *min–max backbone survived*; what changed was scale, evaluation realism, and the surrounding model ecosystem.

## 13 Recent influential conformal works (2023–2025)

Conformal prediction had an unusually strong recent wave of **theory-organizing** work. That matters a great deal for students, because a field can be influential not only through new algorithms but also through new common language and proof templates.

## 13.1 Theory-organizing works that changed how the field is taught

**Recent-work note: The Gentle Introduction and Theoretical Foundations**

**Relation to the canonical framework.** These works do not change the basic set form

$$\widehat{C}(x) = \{y : V(x, y) \le \hat{q}\}.$$

They change how the field is explained.
**Main contribution.** Unified notation, proof patterns, and design choices scattered across many papers.
**Why they mattered.** They lowered the barrier to entry for students and researchers and helped reveal that many conformal variants differ only in the score, the calibration distribution, or the threshold update rule [30, 48].

## 13.2 Recent algorithmic directions

**Recent-work note: Conformal PID control**

**Relation to the canonical framework.** Same conformal set construction, richer online threshold dynamics.
**Key update.** If $g_t = \mathrm{err}_t - \alpha$, then the threshold combines proportional, integral, and predictive terms:

$$q_{t+1} = \underbrace{\eta g_t}_{\mathrm{P}} + \underbrace{r_t\left(\sum_{i=1}^{t} g_i\right)}_{\mathrm{I}} + \underbrace{g_t'}_{\text{D/scorecaster}}.$$

**Claim.** Trend-aware online updates handle seasonality, drift, and temporal structure better than one-bit reactive updates.
**Evidence.** Long-run coverage guarantees and forecasting experiments [31].

**Recent-work note: Online conformal inference under arbitrary distribution shift**

**Relation to the canonical framework.** ACI's idea is preserved, but the theoretical target changes from global long-run behavior to stronger local guarantees over moving windows.
**Key idea.** Update thresholds with online-learning style control so old data do not dominate new drift.
**Claim.** Better local adaptation under arbitrary nonstationarity.
**Evidence.** Regret-style guarantees and online experiments [34].

**Recent-work note: Adapting conformal prediction to distribution shifts without labels**

**Relation to the canonical framework.** Keep the threshold, change the score using an unlabeled target uncertainty statistic.
**Key formula.** If $H(x)$ is predictive entropy on unlabeled target data and $\eta$ an entropy quantile, ECP rescales the score:

$$\widetilde{s}(x, y) = \eta\, s(x, y), \qquad C_{\mathrm{ECP}}(x) = \{y : \widetilde{s}(x, y) \ge q\}.$$

**Claim.** Unlabeled target uncertainty can improve set quality under shift even without target labels.
**Evidence.** Experiments across several datasets and architectures [35].

**Recent-work note: Multi-source conformal inference**

**Relation to the canonical framework.** Same final conformal set, different way to estimate the target quantile.
**Key formula.** Site-specific quantile estimates $\hat{q}_k$ are aggregated as

$$\hat{q}_{\mathrm{fed}} = \sum_{k=0}^{K} a_k \hat{q}_k, \qquad a_k \ge 0, \ \sum_k a_k = 1,$$

where the weights downweight heterogeneous or unhelpful sources.
**Claim.** Multiple imperfect sources can still improve target coverage/efficiency if they are weighted intelligently.
**Evidence.** Efficient-influence-function theory and empirical studies on simulated and clinical data [36].

**Recent-work note: Error-quantified conformal inference (ECI)**

**Relation to the canonical framework.** Same prediction-set form, smoother online threshold updates.
**Key idea.** Replace the binary-feedback update by a gradient step on a smoothed quantile loss:

$$q_{t+1} = q_t - \eta \nabla_q \widetilde{\rho}_\alpha(s_t - q)\Big|_{q=q_t}.$$

**Claim.** Using *how much* a score misses the threshold yields tighter sets than using only hit/miss information.
**Evidence.** Long-run coverage guarantees under dependence and empirical gains on time-series tasks [50].

### 13.3 Why these conformal works mattered broadly

The broad recent conformal story is:

1. the field acquired a much clearer common language and pedagogy;
2. online adaptation under drift became a major research direction;
3. unlabeled-target and multi-source settings pushed conformal prediction closer to real deployment;
4. several methods kept the same final set form but changed the score, the weights, or the threshold dynamics.

This is precisely why conformal prediction is a good teaching case for students: many sophisticated papers are best understood as small but meaningful modifications of one simple backbone.

Part IV

# Part IV: Practical guidance, self-checks, and appendices

## 14 Practical guidance and reporting checklists

### 14.1 If you are deploying a model, which framework should you try first?

**A deployment decision checklist**

1. **Ask whether the label set is still complete.** If not, OOD detection or abstention is needed.
2. **Ask whether the semantic task itself changed.** If yes, simple covariate- or label-shift correction may be invalid.
3. **Ask whether an attacker can choose perturbations.** If yes, ordinary distribution-shift methods are not enough.
4. **Ask whether a single prediction is acceptable.** If not, use conformal prediction or another set-valued uncertainty method.
5. **Ask what data you have at deployment.** Labeled target data, unlabeled target data, multiple source domains, or a stream over time point to different variants.

## 14.2 OOD detection reporting checklist

A trustworthy OOD report should state:

- which score is used (MSP, energy, Mahalanobis, learned score, VLM score, etc.);
- which benchmark slices are used (near-OOD, far-OOD, covariate/full-spectrum);
- how the threshold is chosen;
- at least one threshold-free metric (AUROC/AUPR) and one operating-point metric (for example FPR95);
- whether the OOD benchmark is free of obvious contamination or shortcut artifacts.

## 14.3 Adversarial robustness reporting checklist

A trustworthy robustness report should state:

- norm, budget, and clipping domain;
- attack steps, step size, restarts, and whether AutoAttack is used;
- clean accuracy and robust accuracy;
- whether robustness is evaluated only in-distribution or also under data shift;
- whether the defense may rely on nondifferentiable or stochastic operations that require special evaluation care.

## 14.4 Conformal reliability reporting checklist

A trustworthy conformal report should state:

- the score $V$ being calibrated;
- the calibration split and calibration size;
- target miscoverage level $\alpha$;
- empirical coverage and average set size/interval length;
- coverage under subgroup or shift stratification if relevant;
- the adaptation protocol under shift (weighted, label-shift, ACI, PID, ECP, multi-source, etc.).

### 14.5 Common failure modes across all four areas

| **Failure mode** | What it looks like in practice |
|---|---|
| **Wrong problem framing** | Treating new semantic classes as if they were only a class-prior change, or treating adversarial perturbations as ordinary covariate shift. |
| **Mismatch between theory and data** | Assuming covariate shift even though $p(y \mid x)$ changed, or assuming exchangeability under strong drift with no adaptation. |
| **Benchmark shortcuts** | Detector looks strong because it exploits preprocessing artifacts or trivial far-OOD cues. |
| **Weak evaluation** | Robust model is declared strong under a weak attack; conformal method is declared valid using the same data for calibration and assessment. |
| **Too much faith in a single metric** | High AUROC but poor operating-point performance; correct long-run coverage but unusably large conformal sets. |

## 15 A student self-check: five questions you should be able to answer

After reading Parts I and II, a novice should be able to answer the following without looking back.

1. If $p_t(x)$ changes but $p_t(y \mid x)$ stays the same, why does importance weighting appear?
2. Why is OOD detection usually described as *score + threshold*?
3. Why does FGSM use the sign of the gradient under an $\ell_\infty$ attack?
4. Why does exchangeability imply that conformal ranks are uniformly distributed?
5. Why can a benchmark paper be as influential as an algorithm paper in OOD or robustness research?

A good sign that the beginner path succeeded is that you can answer all five questions in plain English first, and only then with equations.

## 16 Open problems and frontier directions

### 16.1 Distribution shift

Even the classical shift categories do not solve the hardest deployment cases. Important open problems include:

- detecting when covariate-shift or label-shift assumptions are *wrong*;
- adapting under simultaneous shifts where both $p(x)$ and $p(y \mid x)$ change;
- estimating target risk reliably when support overlap is weak.

### 16.2 OOD detection

Important frontier questions include:

- how to separate semantic novelty from benign covariate change in realistic data;
- how to benchmark OOD detection for VLMs and multimodal systems without saturating the task;
- how to obtain calibrated and decision-relevant thresholds when OOD is heterogeneous.

### 16.3 Adversarial robustness

Important frontier questions include:

- how to scale certified defenses to modern foundation-model regimes;
- how to evaluate robustness jointly under natural distribution shift and adversarial shift;
- how to characterize robustness for multimodal and generative systems, not only closed-world classifiers.

### 16.4 Conformal prediction

Important frontier questions include:

- how to obtain useful conditional guarantees rather than only marginal guarantees;
- how to adapt rapidly under strong and structured nonstationarity;
- how to combine multiple sources, privacy constraints, and missing data while keeping the method interpretable for deployment.

## 17 Conclusion

This survey was organized around one simple idea: deployment reliability starts by identifying *what changed*. The four major responses studied here are different because the failure modes are different.

- Distribution shift asks how to estimate or control target risk when the task is still basically the same.
- OOD detection asks whether the input should be rejected because the closed-world class set is no longer adequate.
- Adversarial robustness asks how to protect performance against worst-case perturbations inside a threat model.
- Conformal prediction asks how to return uncertainty sets with formal coverage guarantees, and how those guarantees can be adapted under shift.

For a novice, the most important achievement is not memorizing many papers. It is learning the four canonical skeletons:

$$\begin{aligned}&\text{reweight risk,}\\&\text{score and threshold,}\\&\text{min–max optimization,}\\&\text{score quantile and prediction set.}\end{aligned}$$

Once those skeletons are clear, both the historical roots and the recent leaves become much easier to organize.

## A Appendix A: mathematical mini-primer

### A.1 Bayes' rule and label-shift correction in four lines

Bayes' rule says

$$p(y \mid x) = \frac{p(x \mid y)p(y)}{p(x)}.$$

Under label shift,

$$p_{\mathrm{t}}(x \mid y) = p_{\mathrm{s}}(x \mid y).$$

So

$$p_{\mathrm{t}}(y \mid x) \propto p_{\mathrm{s}}(x \mid y)p_{\mathrm{t}}(y).$$

Replacing $p_{\mathrm{s}}(x \mid y)$ by source Bayes' rule yields

$$p_{\mathrm{t}}(y \mid x) = \frac{r(y)p_{\mathrm{s}}(y \mid x)}{\sum_{y'} r(y')p_{\mathrm{s}}(y' \mid x)}, \qquad r(y) = \frac{p_{\mathrm{t}}(y)}{p_{\mathrm{s}}(y)}.$$

## A.2 Why FGSM uses the sign of the gradient

Under an $\ell_\infty$ budget, each coordinate satisfies $|\delta_i| \le \varepsilon$. The first-order objective is

$$\max_{\|\delta\|_\infty \le \varepsilon} g^\top \delta = \max \sum_i g_i \delta_i.$$

For each coordinate $i$, the best choice is

$$\delta_i = \varepsilon \, \mathrm{sign}(g_i),$$

because that makes $g_i \delta_i = \varepsilon |g_i|$. Therefore

$$\delta^\star = \varepsilon \, \mathrm{sign}(g).$$

## A.3 Why Mahalanobis distance appears in Gaussian feature models

If

$$\phi(x) \mid Y = y \sim \mathcal{N}(\mu_y, \Sigma),$$

then

$$\log p(\phi(x) \mid y) = -\frac{1}{2}(\phi(x) - \mu_y)^\top \Sigma^{-1}(\phi(x) - \mu_y) + \text{constant}.$$

So maximizing Gaussian log-density is the same as minimizing Mahalanobis distance.

## A.4 Why exchangeability gives the conformal rank argument

If $V_1, \ldots, V_n, V_{n+1}$ are exchangeable, then any ordering of these scores is equally likely. Therefore the rank of $V_{n+1}$ among the $n+1$ values is uniform on $\{1, \ldots, n+1\}$ up to ties. Choosing the $k = \lceil (n+1)(1-\alpha) \rceil$-th calibration quantile ensures that the probability the future score lands above that threshold is at most $\alpha$.

## A.5 How a confusion matrix leads to BBSE

Let $\hat{Y}$ be the model prediction and define the source confusion matrix

$$C_{\hat{y},y} = P_{\mathrm{s}}(\hat{Y} = \hat{y} \mid Y = y).$$

If this conditional behavior stays stable under label shift, then

$$q_{\mathrm{t}}(\hat{y}) = P_{\mathrm{t}}(\hat{Y} = \hat{y}) = \sum_y C_{\hat{y},y} \pi_{\mathrm{t}}(y).$$

In vector form,

$$q_{\mathrm{t}} = C\pi_{\mathrm{t}}.$$

So we can recover target priors by solving

$$\pi_{\mathrm{t}} = C^{-1} q_{\mathrm{t}},$$

provided $C$ is invertible and estimated well enough.

# B Appendix B: optional advanced notes

## B.1 Why the Ben-David bound is historically important

A simplified version of the classical adaptation result states that for a hypothesis $h$,

$$\epsilon_{\mathrm{t}}(h) \leq \epsilon_{\mathrm{s}}(h) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(P_{\mathrm{s}}, P_{\mathrm{t}}) + \lambda^{\star},$$

where $d_{\mathcal{H}\Delta\mathcal{H}}$ measures how distinguishable the two domains are using hypotheses in the class, and $\lambda^{\star}$ is the error of the best hypothesis that works on both domains. This bound is historically important because it showed early on that target error depends not only on source accuracy but also on domain mismatch and task compatibility [5].

## B.2 OpenMax in one paragraph

OpenMax starts from activation vectors in the penultimate layer. For each class it estimates a mean activation vector and fits an extreme-value distribution to large distances from that mean. At test time it downweights class scores whose activations fall into the extreme tail and reallocates some mass to an “unknown” class [9]. The conceptual move is to add an explicit unknown option rather than only using a confidence threshold.

## B.3 Why influence functions appear in multi-source conformal

In multi-source problems the target quantile is not observed directly, but it can be expressed as a functional of nuisance quantities such as conditional score distributions and density ratios across sites. Efficient influence functions summarize how that functional changes under small perturbations of the data-generating distribution. They are therefore natural tools for building estimators that combine multiple sources while retaining asymptotic efficiency. A first-time reader does not need the full semiparametric machinery; it is enough to know that the goal is still the same old conformal goal: estimate the right target quantile.

## B.4 One last reading suggestion for beginners

If this document still feels dense, here is a minimal path.

1. Re-read Section 1 and the running example.
2. Read only the summary boxes in Sections 2–5.
3. Work through the toy examples by hand.
4. Read the first derivation in each of Sections 6–9.
5. Use Part III only after you can explain the four skeletons in plain language.

That order is often more effective than reading every page in sequence.

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